

(D) Signal transmission in retina:

- Electrical response of retinal neurons is RP (local graded potential) & not AP → Conduction of visual signals by electronic conduction → allows graded conduction of signal strength

N.B. Exceptions are ganglion cells & some amacrine cells which conduct signals as AP

Lecture (20): Light-Dark Adaptation & Color Vision

(Dr. Wagdy Nashaat Habeeb)

1. Retinal Adaptation (Automatic Regulation of Retinal Sensitivity)

➤ Definition:

Ability of retina to adjust its sensitivity to different light intensities

➤ Types:

	Dark adaptation	Light adaptation
1. <u>Definition:</u>	↑↑ retinal sensitivity to light (by 100.000 to 500.000 times), <i>i.e.</i> , ↓↓ retinal threshold	↓↓ retinal sensitivity to light, <i>i.e.</i> , ↑↑ retinal threshold
2. <u>Occurs when:</u>	Person shifts from brightly-lighted place to dark place	Person shifts from dark place to brightly-lighted place
3. <u>Timing:</u>		
(a) Starts:	After 1 min	After 1 sec
(b) Reaches max.:	Within 30 min	Within 5 min

4. <u>Mechanism:</u>	Regeneration of photopigments 2 stages: *Rapid & small rise: in cones *Slow & greater rise: in rods	Breakdown of photopigments
5. <u>Changes:</u>	(a) Mydriasis (b) Regeneration of photopigments (c) ↑↑ retinal sensitivity to light (d) ↓↓ signal discharge in retinal neurons	(a) Miosis (b) Breakdown of photopigments (c) ↓↓ retinal sensitivity to light (d) ↑↑ signal discharge in retinal neurons

2. Color Vision

➤ Definition:

It is the ability of retina (cones) to differentiate between different wavelengths (colors) of the visible spectrum. Each color has a specific wavelength ranging from 4000 to 7000 Å°.

➤ Characteristics of colors:

1. Hue: = the actual color
 2. Saturation: = degree of freedom from dilution with white (purity of color)
 3. Brightness (luminosity): = degree of intensity of light (measured by N° of photons/unit area/sec)
 4. Primary colors: = red, green & blue
 5. Complementary colors: = pairs of colors when mixed together → sensation of white (e.g., yellow + blue 'lights')
- N.B.* Yellow + blue 'pigments' → green
6. Black: = sensation produced by absence of light

➤ Mechanism:

1. Young-Helmholtz theory (Trichromatic theory):

It states that: “There are 3 types of cones, each one contains a different photopigment & is maximally sensitive to one of the 3 primary colors”

*The 3 cone pigments:

(i) Red sensitive pigment: absorbs light maximally in the yellow portion of the spectrum with a peak wavelength at 565 nm (long wavelength pigment) & its spectrum extends far enough into the long wavelength to sense red

(ii) Green sensitive pigment: absorbs light maximally in the green portion of the spectrum with a peak wavelength at 535 nm (medium wavelength pigment)

(iii) Blue sensitive pigment: absorbs light maximally in the blue-violet portion of the spectrum with a peak wavelength at 445 nm (short wavelength pigment)

*Color sensation depends on the relative frequency of impulses in each of the 3 cones (Equal stimulation of the 3 cones → white color)

*Color perception is a retinal phenomenon, while color translation is a cortical phenomenon

2. Neural mechanism of color vision (The retinex theory):

*Color vision is a combined action of neural activity at several layers of the visual pathway, including the retina & the cerebral cortex

*Color information is carried from retina through 3 types of small ganglion cells (broad band, single opponent & double opponent cells), where each color stimulates some cells & inhibit others → parvo cells of LGB → neurons of blobs (color sensitive neurons) in the 1ry visual cortex (area 17) → visual association cortex (areas 18 & 19) → lingual &

fusiform gyri of occipital cortex (color area)

➤ Color blindness:

*Definition:

Inability to perceive portion of the spectrum or to distinguish between colors

*Causes:

1. Hereditary (X-linked recessive): → ♂: 8 % & ♀: 0.4 %
2. Acquired: due to damage of macula (e.g., macular edema) or optic nerve (e.g., optic neuritis)

*Types of congenital color blindness:

1. Color anomaly (Anomalous trichromatic color vision):

= Functioning 3 cone systems, but 1 of them is weak

- Protanomaly: weak red cone system
- Deuteranomaly: weak green cone system
- Tritanomaly: weak blue cone system

2. Color anopia:

= Absent 1 or more cone systems

(a) Dichromatic color vision: = Absent 1 cone system

- Protanopia: red blindness
- Deuteranopia: green blindness

- Tritanopia: blue blindness

(b) Cone monochromatism: = Absent 2 cone systems

(Colors appear as grades of 1 primary color)

(c) Rod monochromatism: = Absent 3 cone systems

(Colors appear as white, black & shades of gray)

*Tests for color vision:

1. Color matching test
2. Ishihara's chart (hidden figure) test
3. Edridge-Green lantern test

3. Visual Acuity

➤ Definition:

It is the degree of perception of fine details & contours of objects by the eye

➤ **Principle**: "Visual 2-point discrimination"

*Definition: It is the ability to see 2 points with minimal separation as 2 separate points

*Requirements:

1. The 2 images must fall on 2 cones separated by at least 1 unstimulated cone (i.e., retinal distance $> 2 \mu$ as foveal cone diameter = 1.5μ) (Why?)
2. The visual angle (formed by light rays from 2 points at the nodal point) is not < 1 minute

➤ Determination:

1. Landolt chart (broken circle C) (Or) 2. Snellen's chart (alphabetical letters)

*The chart has 7 rows of marks with decreasing size from above downwards which can be read by a normal subject, *i.e.*, forming a visual angle = 1 minute, at distances of 60, 36, 24, 18, 12, 9 & 6 meters respectively

*The subject stands at a distance of 6 meters from the chart (to avoid accommodation)

*Each eye is tested separately for seeing the marks from above downwards

*Visual acuity (VA) is a mathematical fraction (ratio of one's acuity: that of a normal subject)

Numerator = distance of the subject from the chart = 6 m

Denominator = distance at which marks can be read by the normal subject

The normal VA is 6/6 (all marks are seen)

If marks are seen down to the 3rd row only → VA is 6/24, *etc.*

➤ Factors affecting VA:

(A) Internal factors:

1. Age: Sharp VA in children → gradual ↓↓ with age → marked ↓↓ in old age (d.t. presbyopia)
2. Pupillo-constriction: ↑↑ VA (as it ↓↓ spherical & chromatic aberrations)
3. Fovea centralis: → maximal VA (Why?)
4. Diseases of eye: e.g., cataract, glaucoma & retinal detachment → ↓↓ VA
5. Errors of refraction: e.g., myopia & hypermetropia → ↓↓ VA

(B) External (stimulus) factors:

1. Degree of illumination of the chart
2. Degree of contrast between the marks & the background

4. Visual Field

➤ Definition:

It is the maximal area of external world seen by one fixed eye

➤ **Measurement:** By “perimeter”

➤ Normal field:

50° upwards

60° nasally

80° downwards

90° temporally

N.B. It is more extensive for white objects, while it becomes smaller successively when blue, green & red objects are used

➤ Importance of determination:

1. Localization of the site of:

(a) lesion in visual pathway

(b) scotomata (blind spots) in retina

2. Diagnosis of some retinal diseases, e.g., retinitis pigmentosa (loss of the peripheral field before the central one)

5. Binocular Vision

➤ Definition:

It is the ability to use both eyes to see one object at the same time without diplopia

➤ Grades:

1st: Simultaneous perception

2nd: Fusion

3rd: Stereopsis

➤ Causes:

1. The central parts of the 2 visual fields overlap to a great extent
2. The images fall on 2 corresponding points on both retinae → fusion in the visual cortex

Q. What are the corresponding retinal points?

N.B. Diplopia occurs if there is unequal action of EOM, *i.e.*, in squint, as the images do not fall on the corresponding points

➤ Advantages of binocular vision over monocular vision:

1. Wider visual field
2. Better “depth perception”
3. “Stereoscopic vision”: the most important & perhaps the only advantage
4. Retinal defects in one eye, either physiological or pathological, are masked by the normal eye
5. Abnormal refraction of one eye is masked by the normal eye

	Depth perception	Stereopsis (Stereoscopic vision)
*Definition:	Determination of distances of objects from each other or from the observer by the visual system	Differentiation of 3 dimensions of objects
	It is a property of monocular vision, but it becomes more accurate with binocular vision	It is a property of binocular vision

*It depends on:	(A) In monocular vision: 1. <u>Relative size of objects</u>: Far objects are smaller 2. <u>Occlusion</u>: Near object covers a part of far object 3. <u>Distribution of light & shades on surface of objects</u> 4. <u>Shadows cast by object upon its surroundings</u> 5. <u>Aerial perspective</u>: = Effect of atmosphere on colors & details of objects. Far objects appear less clear & acquire a more bluish tinge as the viewing distance ↑↑ 6. <u>Linear perspective</u>: // lines appear to converge with distance 7. <u>Parallax</u>: With head movement, near objects appear to move in the opposite direction & far objects appear to move in the same direction (B) In binocular vision: - <u>Stereopsis</u>	Formation of <u>2 slightly dissimilar images</u> on both retinae (as the 2 eyes are few centimeters apart from each other)
------------------------	---	---